

COMMONWEALTH OF AUSTRALIA

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SENG1050

Web Technologies

Week 4 - CSS

Lecture Outline

- ❑ Cascading Style Sheets (CSS)
- ❑ Structure of a CSS File
- ❑ CSS Selectors
- ❑ Common CCS Properties and Values
- ❑ CSS Box Model

Recall from last week: XML

- ❑ After learning HTML, we have been addressing two problems identified with developing websites using solely HTML. The first problem was to address how to update changing data without having to alter entire websites. The second problem was to address how to specify visual styling to ensure consistency across different browsers.
- ❑ Both problems have the same root, and the same solution, we need to separate the structure of the website from the data and from the visual styling.
- ❑ We have learned to separate the data from the structure using XML with a DTD to store our website data. This means we now have only one file to update when data changes instead of multiple files.
- ❑ We now turn our attention to the second problem, specifying visual styling across our webpage, both to separate it from the structure of the website (as with the data), and to ensure our website looks the same across all media.

Cascading Style Sheets (CSS)

Overview of Client-based Web Development



Visual Formatting

- ❑ HTML tags imply suggestions to the browser about how to draw their content
 - Headings change the font size and weight
 - `<blockquote>` indents the margins
- ❑ These are only suggestions!
 - A browser can ignore them
 - It may not even support them – e.g., some older or simpler mobile phones

```

<!DOCTYPE html>
<html>
  <head>
    <title>An Example</title>
    <meta charset="UTF-8">
  </head>

  <body>
    <h1>An Example</h1>

    <hr>

    <h2>Chapter 1</h2>

    <blockquote><p>
      "Cascading style sheets (CSS) is a stylesheet language used to describe
      the presentation of a document written in a markup language."</p>
    </blockquote>

    <p>Removal of absolute identifier is not enough for privacy protection.</p>

    <p>An intruder having enough supplementary knowledge can re-identify someone in whom
    he/she is interested. Therefore we add noise to a data set and perturb the data
    set.</p>

    <ul>
      <li>Plagiarism is bad</li>
      <li>Poor referencing can be mistaken for plagiarism</li>
      <li>Therefore poor referencing is bad!</li>
    </ul>

    <ol>
      <li>This is an ordered list</li>
      <li>Do you prefer it to the unordered one?</li>
      <li>Perhpas they both have their place</li>
    </ol>
  </body>
</html>

```

An Example

Chapter 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

Removal of absolute identifier is not enough for privacy protection.

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1. This is an ordered list
2. Do you prefer it to the unordered one?
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Semantics!

- ❏ All HTML tags, except a few such as `<div>` and `<span>`, have implicit semantic meanings.
 - For example: `<h1>`, `<p>`, `<strong>`, `<em>`.
 - Before using a tag, think about whether it is really what you “mean” in the given context.
- ❏ Never use a tag just because it gives you the desired visual formatting!!
 - It won’t always work – especially on every platform.
 - There are much better ways to do it.

Cascading Style Sheets (CSS)

- ❑ Style sheets are a way of adding visual formatting to HTML tags
 - CSS is the most common style sheet language
 - CSS is independent of the semantics of the HTML tags – you can change the (suggested) formatting without affecting the semantics!



CSS - Advantages

- ❑ Gives greater control over page layout and typography.
- ❑ Style is separate from structure.
- ❑ Style can be stored separately.
 - Saves work if we want to change many pages.
 - Potentially smaller documents.
 - Site maintenance is easier.

Structure of a CSS File



CSS - Types

- ❏ CSS can be applied to HTML in three main ways
 1. Inline styles
 2. Embedded (Document-level) styles
 3. External style files



General Components

- ❑ Style sheets are composed of rules that describe the styling to be applied.
- ❑ Rules have the following syntax:
selector {property: value}
- ❑ The following CSS rule would set the colour of the text to blue

body {color: blue}

selector

property

value

```
/* stylesheet.css
 * Purpose: to introduce a CSS file
 */
body {
    background-image: url('./background.jpg');
    margin-left: 1cm;
}
h1 {
    color: red;
    text-align: center;
}
h2 {
    color: blue;
    text-align: left;
}
p {
    text-align: left;
    color: black;
    font-family: arial, serif;
    text-indent: +2%;
}
```



Inline Styles

- ❏ Almost every tag in HTML has an optional **style** attribute

`style="property1: value1; ... propertyn: valuen"`

- ❏ The style applies to the content of the tag ... including nested tags

Examples:

```
<h1 style="color:red">Little Red Riding Hood</h1>
```

```
<h1 style="text-align:right">Teach Me Chinese</h1>
```

```
<body style="background-color:green">
```

```
<h1 style="text-align:center">It Ain't Easy...</h1>
```

```

```

```
<!DOCTYPE html>
<html>
  <head>
    <title>An Example</title>
    <meta charset="UTF-8">
  </head>

  <body>
    <h1>An Example</h1>

    <hr>

    <h2>Chapter 1</h2>

    <blockquote><p>
      "Cascading style sheets (CSS) is a stylesheet language
      the presentation of a document written in a markup lang
    </blockquote>

    <p style="color:red; font-weight:bold">
      Removal of absolute identifier is not enough for privacy
```

An Example

Chapter 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

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Embedded Styles

```
<head>
  <style>
    selector { property1: value1; ...
  }
  ...
</style>
</head>
```

- ❑ **selector** determines which tags the style applies to.
- ❑ Applies to all such tags in the current document.

```
<style>
  body { color: white; background-color: green;
        text-align: left }

  h1 { color: yellow; text-align: center }
  h2 { font-style: italic }
  strong { font-size: larger; color: red }
  img { float: right }
  img#special { width: 150px; height: 200px }
  #green {color:green}
</style>
```

```

<!DOCTYPE html>
<html>
  <head>
    <title>An Example</title>
    <meta charset="UTF-8">
    <link rel="stylesheet" type="text/css" href="style.css" />

    <style>
      blockquote {background-color: yellow}
      hr {border: 1px dashed red}
    </style>
  </head>
  <body>
    <h1>An Example</h1>

    <hr>

    <h2>Chapter 1</h2>

    <blockquote><p>
      "Cascading style sheets (CSS) is a stylesheet language us
      the presentation of a document written in a markup langua
    </blockquote>

```

An Example

Chapter 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

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External Style File

<head>

```
<link rel="stylesheet" type="text/css"
href="uri" />
```

</head>

- ❑ Applies the CSS rules found in the external style file *uri* to this document.
- ❑ The style file should contain only rules of the form selector { property : value ; ... } – **no <style> or <head>**.
- ❑ Comments in the style file have the form /* ... */.

```
/* stylesheet.css
 * Purpose: to introduce a CSS file
 */

body {
    background-image: url('./background.jpg');
    margin-left: 1cm;
}

h1 {
    color: red;
    text-align: center;
}

h2 {
    color: blue;
    text-align: left;
}

p {
    text-align: left;
    color: black;
    font-family: arial, serif;
    text-indent: +2%;
}
```

```

/* stylesheet.css
 * Purpose: to introduce a CSS file
 */

body {
    background-image: url('./background.jpg');
    margin-left: 1cm;
}

h1 {
    color: red;
    text-align: center;
}

h2 {
    color: blue;
    text-align: left;
}

p {
    text-align: left;
    color: black;
    font-family: arial, serif;
    text-indent: +2%;
}

```

```

<html>
  <head>
    <title>An Example</title>
    <meta charset="UTF-8">
    <link rel="stylesheet" type="text/css" href="stylesheet.css">
  </head>

```

An Example

Chapter 1

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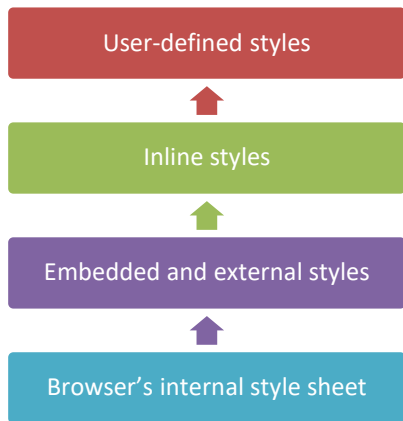
- Plagiarism is bad
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CSS Cascading Order

Generally speaking, we can say that all the styles will "cascade" into a new "virtual" style sheet by the following rules, where the top has the highest priority:



- ❑ Browser default style rules are overridden by rules in an external style file or document-level style tag (applied in order, so the last style applied takes precedence)
- ❑ Rules in an external style file or document-level style tag are overridden by rules in an *in-line* style attribute
- ❑ So **inline style rules have the highest priority**
- ❑ Except for user defined instructions to the browser.



Using Cascading Style Sheets

- ☐ Place style rules that apply to your **entire website** in an external style file and link all documents to it
- ☐ Place styles rules that apply only to **a group of documents** in an external style file and link only those documents to it (you can have many `<link>` tags).
- ☐ Place styles particular to **a given document** in a embedded (document-level) style tag.
- ☐ Only use inline styles for **one-off visual formatting** if you are sure it is not going to be re-used – even then, a document-level rules is probably better.
- ☐ Use class attributes to distinguish tags with different semantic meanings – even if the visual formatting is the same.

CSS Drawbacks

- ❑ CSS is only a suggestion to the browser – it might be ignored.
- ❑ Not every property applies to every tag.
- ❑ Not every browser supports every property, or every value for a property.
- ❑ Users can override any or all CSS rules at the browser level.

Checking Browser Support

- ❏ CanIUse.com
- ❏ They test almost every CSS feature and the test the compatibility a long way back, version-wise, and on every mainstream browser.



Validating CSS

- ❑ Mistakes in CSS code can be hard to pickup.
 - Many browsers will stop processing a CSS rule if an error is discovered.
 - Some will stop processing an entire style (document-level tag or external link).
- ❑ Or your CSS might be correct, but the browser might not support the property or value.
- ❑ Use a CSS validator
 - <http://jigsaw.w3.org/css-validator/>
 - May fail if the HTML is invalid.

CSS Debugging Tips

- ❑ Manually check syntax errors.
- ❑ Use W3C CSS Validator to check syntax errors.
 - <http://jigsaw.w3.org/css-validator/>
- ❑ Configure temporary background colors.
- ❑ Configure temporary borders.
- ❑ Use CSS comments to find the unexpected:
`/* the browser ignores this code */`
- ❑ Be patient!

CSS Selectors



CSS Selectors

```
selector {property1: value1; property2: value2; ...}
```

selector → h1 {color: blue; font-size: 12px;}

property ↑

value ↑



CSS Selectors

selector can be

- ❑ A tag name
`p { text-align: justify }`
- ❑ Multiple tags (or even other selectors)
`p,blockquote { text-align: justify }`
- ❑ A tag descended (child) from another tag
`div p { font-weight: bold }`
- ❑ A tag that is an immediate child of another tag
`div > p { background-color: yellow }`
- ❑ A tag that immediately follows another tag
`div + p { background-color: yellow }`
- ❑ A tag that follows another tag
`div ~ p { background-color: yellow }`

```

<body>
<h1>Welcome to My Homepage</h1>

<div>
  <h2>My name is Donald</h2>
  <p>I live in Duckburg.</p>
</div>

<div>
  <span><p>I am not a direct child.</p></span>
</div>

<p>My best friend is Mickey.</p>
<h2>The End</h2>
</body>

```

```

div > p {
  background-color: yellow;
}

div + p {
  border: 1px solid red;
}

div p {
  text-transform: uppercase;
}

div ~ h2 {
  text-align: center;
}

```

This rule applies to the **second h2 (The End)** because it is preceded by a **div**.

It won't apply to the **first h2 (My name ...)** because it appears before a **div** element has ended.

Welcome to My Homepage

My name is Donald

I LIVE IN DUCKBURG.

I AM NOT A DIRECT CHILD.

My best friend is Mickey.

The End

Both these p's
appear inside a div
(not necessarily
immediate children)

CSS Selectors

selector can be

- ❑ A tag that is the nth child of its parent
`li:nth-child(1) { font-size: 110% }`
- ❑ Every nth tag of its parent
`li:nth-child(2n) {background-color: grey }`
`li:nth-child(even) {background-color: grey }`
- ❑ A tag that is the nth last child of its parent
`h2:nth-last-child(3) {text-decoration: underline }`
- ❑ A tag that the nth tag of its type, of its parent
`p:nth-of-type(3) {color: green}`
- ❑ A tag that is the nth last tag of its type, of its parent
`p:nth-last-of-type(2) {color: yellow}`
- ❑ A tag that is the only child of its parent
`li:only-child {font-weight: bold}`



CSS Selectors

*(pseudo-)*selector can be used for:

- ❑ A hyperlink **that has not been visited yet**
`a:link { color: blue }`
- ❑ A hyperlink that has already been visited
`a:visited { color: black }`
- ❑ A tag during a user action
`a:active { color: red }`
`a:hover { font-size: 120% }`
- ❑ Not just links!
`div:hover { color: yellow }`

```

<address>
  <a href="https://www.newcastle.edu.au/profile/raymond-chiong">
    Raymond Chiong
  </a><br>
  (02) 4921 7367<br>
  School of Information and Physical Sciences<br>
  University of Newcastle
</address>

</body>
</html>

```

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[Raymond Chiong](#)

(02) 4921 7367

School of Information and Physical Sciences

University of Newcastle

```

address { text-align: right }

a:link { color: blue }
a:visited { color: green }
a:active { color: red }
a:hover { font-size: 120%; color: purple }

```

[Raymond Chiong](#)
 (02) 4921 7367
 School of Information and Physical Sciences
 University of Newcastle

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CSS Selectors

selector can be...

- ❑ A tag with a given attribute

```
input[checked] { color: red }
```

- ❑ A tag with a given attribute with an exact given value

```
a[href="home.html"] { color: green }
```

- ❑ A tag with a given attribute containing a given substring

```
a[alt*="flower"] { border: 1px solid blue }
```

- ❑ A tag with a given attribute starting with a given value

```
a[href^="https://en.wikipedia"] { font-weight: bold }
```



A tag with a given **id** attribute value **[ID Selector]**

```
#copyright { font-style: italic }
```



The Class Attribute

There is a special attribute for use when formatting HTML using CSS – **class**.

- ❑ It contains a space-separated list of “class names”.
- ❑ Classes can be used in CSS selectors.

`<div class="copyright">...</div>` with
`div.copyright { font-style: italic }`

- ❑ The class attribute is also very useful for associating extra semantic meaning with HTML tags
`p.intro, p.body, p.conclusion { ... }`

```
<p style="color:red; font-weight:bold" class="right">  
Removal of absolute identifier is not enough for privacy protection.  
</p>
```

```
p.right {  
  text-transform: uppercase;  
  text-align: right  
}
```

An Example

Chapter 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

REMOVAL OF ABSOLUTE IDENTIFIER IS NOT ENOUGH FOR PRIVACY PROTECTION.

An intruder having enough supplementary knowledge can re-identify someone in whom he/she is interested. Therefore we add noise to a data set and perturb the data set.



 and <div>

Most HTML tags have implicit visual formatting

- ❑ CSS lets you override these, but you can get caught if a particular browser uses an unusual property – e.g., Lynx underlines `<em>` text; if you don't set `text-decoration` in the CSS style, the underline will not be overridden.
- ❑ Only `<div>` and `<span>` are guaranteed not to carry implicit visual formatting, so you should use these if another tag is not required for its semantics.
- ❑ You can add semantics to `<div>` and `<span>` using classes.



CSS Specificity

What happens when two rules conflict?

- ❑ Example: for a HTML element `<div class="bar" id="foo"></div>`

```
#foo {color: red}  
.bar {color: green}
```

Which rule will be applied?

- ❑ When multiple rules in the CSS cascade order conflict, the most specific relevant selector is used to determine which rule to apply.
- ❑ Specificity is made up of three categories, from most to least specific - IDs, classes, and then types (including pseudo-elements). The number of IDs in each relevant selector is checked, if there is a tie, then the number of classes in each relevant selector is checked, and then finally the number of types.
- ❑ In the above example, the first rule would be applied because IDs are more specific than classes.
- ❑ If the specificity score is still tied, the order is used, with the last relevant selector being applied.

An advanced example of specificity:

```
#b, #b.c, a.c.d.e {color: red}  
.a, #b.d, a.b#b:link {color: blue}
```

What will be applied to the HTML element:

```
<a class="a b c d e", id="b" href="...",  
alt="...">...</a>
```

The most specific selector is "a.b#b:link", so this rule will be applied.

Consider the following HTML file, what colour will line 3 be?

Red

Yellow

Green

Blue

Black



[PollEv.com/adriantan744](https://pollev.com/adriantan744)

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Document</title>
  <style>
    div p {color:red}
    div > p {color:yellow}
    div ~ p {color:green}
    div + p {color:blue}
  </style>
</head>
<body>
  <div>
    <p>Line 1</p>
    <p>Line 2</p>
    <section>
      <p>Line 3</p>
    </section>
    <p>Line 4</p>
    <p>Line 5</p>
  </div>
  <p>Line 6</p>
  <p>Line 7</p>
</body>
</html>
```

0% 0% 0% 0% 0%

Consider the following HTML file, what colour will the paragraph be?

Red

0%

Yellow

0%

Green

0%

Blue

0%

Purple

0%



PollEv.com/adriantan744

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Document</title>
  <style>
    p {color: red}
    .a {color: yellow}
    #b {color: green}
    p.b.c {color: blue}
    #a {color: purple}
  </style>
</head>
<body>
  <p class="a b c" id="a">Hello World!</p>
</body>
</html>
```

Common CCS Properties and Values

CSS Properties and Values

- ❑ Here are some of the most common CSS properties and their values
 - There are many, many more
- ❑ For the gory details, see <https://www.w3.org/TR/css3-roadmap/>
- ❑ Tutorials
 - http://www.w3schools.com/css/css_intro.asp
 - https://developer.mozilla.org/en-US/docs/Web/Guide/CSS/Getting_started



CSS Properties and Values

font-size

- ❑ Relative: `xx-small` | `x-small` | `small` | `medium` | `large` | `x-large` | `xx-large`
- ❑ Relative: `smaller` | `larger` | `%` | `em` | `rem` | `vw` | `vh`
- ❑ Absolute: `cm` | `mm` | `in` | `pt` | `pc` | `em` | `ex` | `px`



CSS - Units

- ❑ CSS understands multiple types of units for expressing sizes and quantities.
- ❑ Units are divided into two types, absolute and relative.
- ❑ Absolute units are fixed and are relative only to the actual size of the screen. For example:
 - ❑ px (pixels – 1/96 of an inch)
 - ❑ pt (points – 1/72 of an inch)
 - ❑ cm (centimetres)
 - ❑ in (inches)
- ❑ Relative units are relative to some other property of the webpage or window. For example:
 - ❑ % (percentage relative to the parent element – 100% would mean no change)
 - ❑ em (relative to the font-size of the parent element – 1em would mean no change)
 - ❑ rem (relative to the font-size of the root element)
 - ❑ vw (relative to the window width in % – 100vw would be the entire window width)
 - ❑ vh (relative to the window height in % - 100vh would be the entire window height)



Fonts

`font-family`

- ❑ A comma-separated list of fonts, in order of preference.
- ❑ Names with spaces must be in double- or single-quotes.
- ❑ You should include at least one of these: `serif`, `sans-serif`, `cursive`, `fantasy`, `monospace`.
- ❑ http://en.wikipedia.org/wiki/Font_family_%28HTML%29



Fonts

- ❑ **font-style:** `normal` | `italic` | `oblique`

```
p.normal {font-style:normal}  
p.italic {font-style:italic}  
p.oblique {font-style:oblique}
```

- ❑ **font-variant:** `normal` | `small-caps`

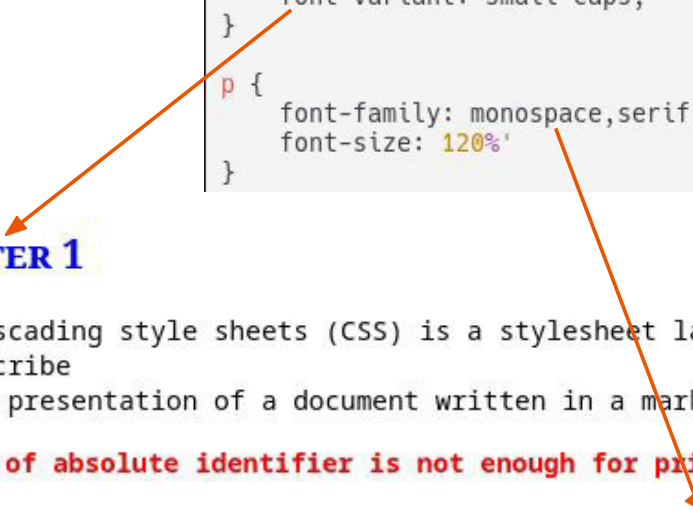
```
p.small {font-variant: small-caps}
```

- ❑ **font-weight:**

`normal` | `bold` | `100` | ... | `900`
`lighter` | `bolder`

```
p.normal {font-weight: normal}  
p.thick {font-weight: bold}  
p.thicket {font-weight: 900}
```

```
h2 {  
  color: blue;  
  font-variant: small-caps;  
}  
  
p {  
  font-family: monospace, serif;  
  font-size: 120%;  
}
```

Two orange arrows originate from the CSS code block. One arrow points from the 'h2' selector to the 'CHAPTER 1' title. The other arrow points from the '120%' value in the 'font-size' property to the word 'privacy' in the sentence 'Removal of absolute identifier is not enough for privacy protection.'

CHAPTER 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

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Fonts

- ❏ `text-decoration:`
 - `none | overline | underline | line-through`
- ❏ `text-align:`
 - Horizontal alignment of text
`left | right | center | justify`
- ❏ `text-indent:`
 - Indent (+ or -) for the first line
 - A length or percentage

- ❏ `text-shadow:`
 - Specify the horizontal shadow, the blur distance, and colour of the shadow
`h-shadow v-shadow blur-color | none | initial | inherit;`

```
h1 {text-shadow:2px 2px  
#FF0000;}
```

An Example



Fonts

- ❑ `text-transform:`
 - `capitalize` | `uppercase` | `lowercase` | `none` | `inherit`
- ❑ `white-space:`
 - `normal` | `pre` | `nowrap`

white-space: nowrap

```
<p>I have      multiple  
spaces and a line break</p>
```

WILL LOOK LIKE



I have multiple spaces and a line break

(multiple spaces ignored, and line breaks will never occur, even if space is limited)



Lists

❑ `list-style-type`:

- change the behaviour of list markers
- `disc` | `square` | `circle` – unordered markers
- `decimal` | `lower-roman` | `upper-roman` | `lower-alpha` | `upper-alpha` | ...

❑ `list-style-image: url(uri)`

```
ul { list-style-type: square }  
ol { list-style-type: lower-roman }
```

- Plagiarism is bad
 - Poor referencing can be mistaken for plagiarism
 - Therefore poor referencing is bad!
- i. This is an ordered list
 - ii. Do you prefer it to the unordered one?
 - iii. Perhaps they both have their place



Background

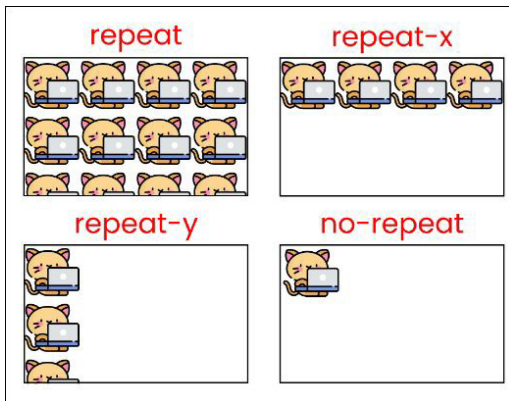
- ❑ *background-color*: Sets the background color of an element.
- ❑ *background-image*: Sets one or several background images for an element.
- ❑ *background-repeat*: Specifies how background images are tiled after they have been sized and positioned.
- ❑ *background-clip*: Specifies the area within which the background is painted.
- ❑ *background-position*: Sets the initial position of the background image.
- ❑ *background-size*: Specifies the size of the background images.
- ❑ *background-attachment*: Specifies whether the background scrolls with the document, or remains fixed to the viewing area.
- ❑ *background-origin*: Specifies the positioning area of the background images

`background: [ image position/size repeat attachment origin clip color ]`

Source: <https://www.tutorialrepublic.com/css-reference/css-background-property.php>

Background

❏ background-repeat: repeat | repeat-x | repeat-y | no-repeat | space | round

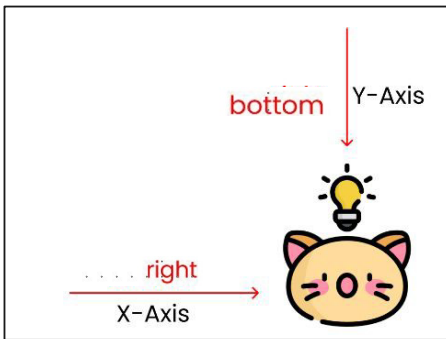


Source: <https://www.freecodecamp.org/news/learn-css-background-properties/>

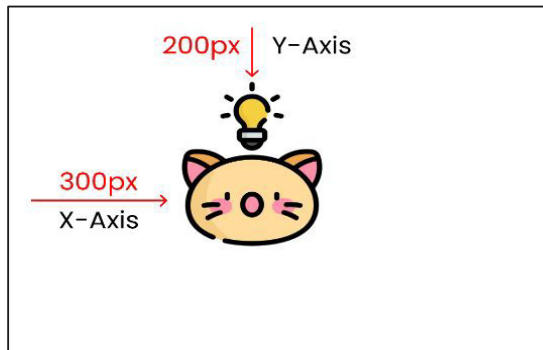
Background

- ❑ `background-position`: combination of `top` | `left` | `right` | `bottom` or exact/percentage values

```
// This is      Y-Axis & X-Axis  
background-position : bottom right;
```



```
// This is      X-Axis & Y-Axis  
background-position : 300px 200px;
```



Source: <https://www.freecodecamp.org/news/learn-css-background-properties/>



Colour

- ❑ Any colour can be thought of as a combination of three primary colours: **red, green, and blue**
- ❑ By varying the intensity of each primary colour, you can create almost any colour and any shade of colour.
- ❑ This principle allows a computer monitor to combine pixels of red, green, and blue to create the array of colours you see on your screen.
- ❑ The intensity of each of three colours (RGB) is assigned a number from 0 (absence of colour) to 255 (highest intensity).
- ❑ In this way, **256^3 , or more than 16.7 million, distinct colours can be defined.**

Current Number: 0

Colour

- ❑ HTML colour values can be entered as hexadecimals.
- ❑ A **hexadecimal** is a number based on a base-16 numbering system rather than a base-10 numbering system that we use every day.
 - Base 10 counting uses a combination of 10 characters (0 through 9) to represent
 - Hexadecimals include six extra characters: A (10), B (11), C (12), D (13), E (14), and F (15).
- ❑ Every colour can be represented as a string of 6 hexadecimal characters – two for red, two for green, and two for blue. For example: #FF0000 indicates the maximum amount of red, with no green, and no blue.
- ❑ With two hexadecimal characters, the highest number we can make – FF, translates into 255.
- ❑ Optionally, two more hexadecimal characters can be added to specify the alpha – i.e. the transparency.

Note:

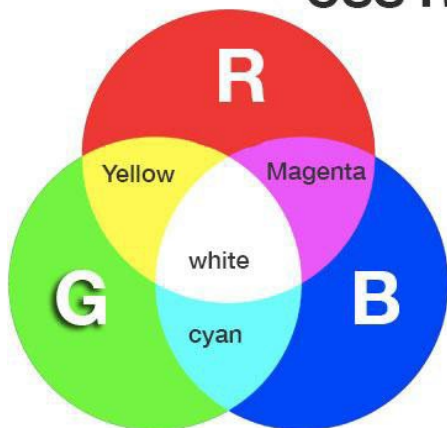
The easiest way to get hex colour codes is to use a colour picker tool. For example see <https://html-color.codes/>

Decimal:	10 ⁴	10 ³	10 ²	10 ¹	10 ⁰
	0	0	0	0	0
Hexadecimal:	16 ⁴	16 ³	16 ²	16 ¹	16 ⁰
	0	0	0	0	0



Colour

CSS RGB COLORS



RED : #FF0000 or `rgb(255,0,0)`

GREEN : #00FF00 or `rgb(0,255,0)`

BLUE : #0000FF or `rgb(0,0,255)`

tutorial.techaltum.com

Using YELLOW that is 25% transparent: #FFFF00BF or `rgba(255,255,0,191)`.

Basic 17 colours for the web

maroon #800000	red #ff0000	orange #ffa500	yellow #ffff00	olive #808000
purple #800080	fuchsia #ff00ff	white #ffffff	lime #00ff00	green #008000
navy #000080	blue #0000ff	aqua #00ffff	teal #008080	
black #000000	silver #c0c0c0	gray #808080		

Source: WebFX

Colour

- ❑ Using the basic colour names allows you to accurately display them across different browsers and operating systems.
- ❑ The list of only 17 colours is limiting to web.
- ❑ CSS3 and almost all browsers support extended colour keywords that you can use in addition to the basic 17 colour names.
- ❑ For a full list of these colour keywords see https://www.w3.org/wiki/CSS3/Color/Extended_color_keywords

Partial list of extended colour keywords

palegoldenrod	#EEE8AA	
palegreen	#98FD98	
paleturquoise	#AFEEEE	
palevioletred	#DB7093	
papayawhip	#FFEFD5	
peachpuff	#FFDAB9	
peru	#CD853F	
pink	#FFC0CD	
plum	#DDA0DD	
powderblue	#B0E0E6	
purple	#800080	
red	#FF0000	
rosybrown	#BC8F8F	
royalblue	#4169E1	
saddlebrown	#8B4513	

Consider the following HTML file, what font size will the paragraph element have?

6pt

0%

9pt

0%

20pt

```
<html lang="en" style="font-size: 20pt">
```

0%

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Document</title>
```

```
  <style>
```

```
    div {font-size: 1.5em}
```

```
    p {font-size: 20%}
```

```
    div p {font-size: 1.5rem}
```

```
  </style>
```

```
</head>
```

```
<body>
```

```
  <div>
```

```
    <div>
```

```
      <p>Hello World!</p>
```

```
    </div>
```

```
  </div>
```

```
</body>
```

```
</html>
```

0%

0%



Which of the following colour values specify a visible red? Select the TWO correct answers.

`rgba(255, 0, 0, 0)`

0%

`#ff000000`

0%

`red`

0%

`#ff0000ff`

0%

`#ff00ffff`

0%

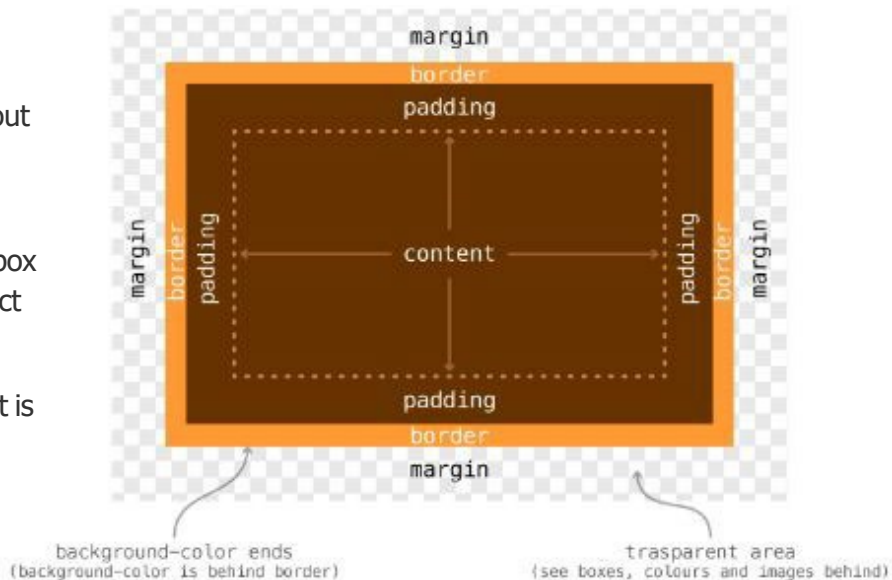


CSS Box Model



CSS Box Model

- ❑ The box model is the name of part of the layout system browsers use when rendering your website.
- ❑ It describes how each of the properties of a box interact with each other and how they interact with other boxes.
- ❑ A box with no content isn't visible, so content is the first layer.



Source: <https://learn-the-web.algonquindesign.ca/topics/box-model/>



Margin & Padding

- ❑ **margin** — adds spacing outside the border area
- ❑ **padding** — adds spacing between the content area and the border area
- ❑ Since boxes have four sides, there are four different margins and paddings that can be changed.
- ❑ **Top and bottom margins can 'collapse' when elements are vertically adjacent**, an element is empty, or a parent element has nothing separating it from its child elements.

- ❑ **margin:** margin-top, margin-left, margin-bottom, margin-right
- ❑ **padding:** padding-top, padding-left, padding-bottom, padding-right

Margin & padding shorthand

- ❑ Instead of using the properties listed above, often it's easier to use the shorthand version.

```
div {  
  /* margin: top right bottom left; */  
  margin: 10px 12px 8px 6px;  
}
```

Source: <https://learn-the-web.algonquindesign.ca/topics/box-model/>



Box Dimension

- ❑ Both **width** and **height** can be assigned to boxes.
 - ❑ Usually we want widths set in some sort of relative measurement, like **%** or **em**.
 - ❑ Try to avoid using height
 - Let the content inside the box dictate the height.
 - ❑ Both width and height apply only to the content area, and excludes the margin, border or padding areas (in other words, adding these things will grow the element box. If you'd prefer your width and height declarations to include borders and padding (more intuitive in some cases), you should set **box-sizing: border-box** which will cause these to be included in the width and height calculation.
- ❑ You can specify the minimum height/width with **min-height** | **min-width** and the maximum height/width with **max-height** | **max-width**

```
width: 100px;  
width: 50%; /* of the element its inside */  
width: 100vw; /* Full width of the window */
```

Source: <https://learn-the-web.algonquindesign.ca/topics/box-model/>

Box Decoration

- ❑ **background-color**: defines the background colour of the content, padding and border areas.
- ❑ **border**: draws a line (border) around the padding area (in the border area).
- ❑ Border can be fine-tuned with related properties
 - **border-width, border-color, border-style**

- ❑ **border-style**: **solid | dashed | dotted | double | groove | ridge | inset | outset**
- ❑ You can also target each side of the box using **border-top, border-left, border-bottom or border-right**.

border shorthand

A short form version for setting the three border properties.

border: border-width border-style border-color

Source: <https://learn-the-web.algonquindesign.ca/topics/box-model/>

Chapter 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

Removal of absolute identifier is not enough for privacy protection.

```
blockquote {  
  width: 80%;  
  background-color: #ffff00;  
  border: 4px dashed green;  
  padding-left: 20px;  
  margin-left: 20px;  
  border-left-color: red;  
}
```

Chapter 1

"Cascading style sheets (CSS) is a stylesheet language used to describe the presentation of a document written in a markup language."

Removal of absolute identifier is not enough for privacy protection.

Images

- ❑ The default width and height of an image can be overridden
 - ❑ `width: Length | auto`
 - ❑ `height: Length | auto`
 - ❑ The image is “scaled” to the new size
 - ❑ If only one is specified (the other is auto) then the aspect ratio is maintained

Images

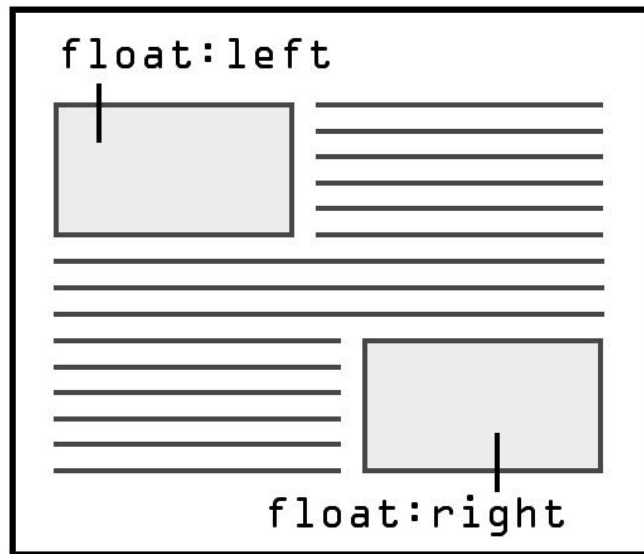
- ❑ Loading a large image then scaling it to a small icon is very inefficient. There are many ways of dealing with this, but one simple process can be:
 - ❑ Generate the small image offline.
 - ❑ Embed the small image in the document with `<img>`.
 - ❑ Hyperlink the small image to the large image file by wrapping `<a>` around `<img>`.
 - ❑ The large image is then only loaded if needed.
- ❑ There are even better ways of handling large images, we will look at these next week.



Images

float:

- ❑ Where to position the image relative to text.
- ❑ **left** – align with the left edge of the enclosing box and text flows around to the right.
- ❑ **right** – align with the right edge of the enclosing box and text flows around to the left.
- ❑ **none** – place inline, or in a new block with text above and below.
- ❑ Works for other content as well.



Source:

<https://pasja-informatyki.pl/programowanie-webowe/metoda-float-left-clear-both/>

<p>
An intruder having enough supplementary info can re-identify someone perturb the data set.

An intruder having enough supplementary knowledge can re-identify someone in whom he/she is interested. Therefore we add noise to a data set and perturb the data set.
An intruder having enough knowledge can re-identify someone in whom he/she is interested. Therefore we add noise to a data set and
</p>

An intruder having enough supplementary info can re-identify someone perturb the data set. An intruder having enough supplementary knowledge can re-identify someone in whom he/she is interested. Therefore we add noise to a data set and perturb the data set. An intruder having enough knowledge can re-identify someone in whom he/she is interested. Therefore we add noise to a data set and



Consider the following HTML files, which CSS rule is most appropriate to add 3 line breaks of spacing between two div elements?

`div {padding: 3em}`

0%

`#a {padding-bottom: 3em}`

0%

`div {margin: 3em}`

0%

`#b {margin-top: 3em}`

0%

None of the above

0%

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Document</title>
</head>
<body>
  <div id="a"></div>
  <div id="b"></div>
</body>
</html>
```



How large will the vertical margin be between 'Line 1' and 'Line 2' given the CSS rule 'p {margin-top: 5px; margin-bottom: 5px; margin-right: 5px; margin-left: 5px}'?

5px

0%

10px

0%

15px

0%

20px

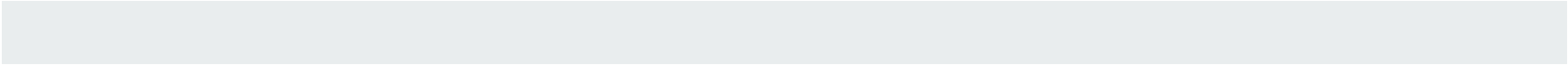
0%

None of the above

0%

```
<body>
  <p>Line 1</p>
  <p>Line 2</p>
</body>
```





Summary - Creating our website

- ❑ We have now addressed the issue of separating the visual styling of our website from the structure and the data. This means we now have only one file to update when the visual styling changes instead of multiple files.
- ❑ The styles that we specify using CSS are applied in every browser generally the same way, ensuring consistency for those styles.
- ❑ There are still problems. What about the styles we don't specify? The browser defaults still apply, and may still differ. What about applying layouts in CSS, such as navigation bars, sidebars, columns, and elements that change to fit the size of the screen? How do we specify the structural sections of a webpage semantically, and style them as a layout? What about more complex elements of a webpage, such as forms, or tables?
- ❑ For simplicity, we initially skipped over these advanced parts of HTML and CSS. Now though, with an understanding of the fundamentals, we can go back and add them to our toolbox – next week.

Summary

- ❑ How different CSS selectors can be used
 - Element Selector
 - ID Selector
 - Attribute Selector
 - Class Selector
- ❑ Common CSS properties and values
 - Font
 - Text
 - Background
 - Margin
 - Border
- ❑ RGB Color scheme
- ❑ How CSS can separate semantics and visual formatting
- ❑ CSS facilitates smart, effective and efficient management, of sites
- ❑ Syntax of CSS
- ❑ Three levels of CSS: inline, embedded and external
- ❑ Cascading order of CSS